

PWM-Correlated Reconstruction and Active Compensation of CM Disturbance Pulses in a Motor Inverter

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1 Introduction

The electrification of drivetrain systems in applications like electric vehicles and aircrafts calls for ever increasing power density and efficiency of the associated motor inverters. However, with the fast switching speed and resulting EMI of modern WBG semiconductors, passive filters can heavily impact overall power density, requiring up to 30 % of overall system volume [1]. Active EMI filters (AEF) achieve strong suppression while maintaining high power density [1].

Common mode (CM) EMI consists of pulses caused by fast voltage slopes and parasitic capacitances in the motor inverter. Hence, CM AEFs are investigated which inject fitting cancellation pulses through various means. Approaches include attaching an additional half-bridge switching complementary to the inverter [2], yielding 25 dB suppression at 160 kHz and 11 dB at 900 kHz, or using a balanced inverter with dual windings [3], achieving 15 dB between 150 kHz and 10 MHz. Alternatively, active circuits with a transformer coupling to the AC lines can provide 20 dB at 100 kHz and up to 8 dB at 10 MHz [4]. The general idea of injecting cancellation pulses can be optimized by tuning the active EMI filter to closely imitate the CM emissions of the drive train. This is done in [5] where a passive small-signal circuit is matched to the measured CM impedance of the system, producing a fitting cancellation signal and achieving high reduction of 33 dB at 150 kHz and 15 dB at 1 MHz.

With a digital AEF (DAEF), the inverter CM EMI pulses can be imitated with a digital cancellation predictor, as generally described in [6] and shown in Figure 1. The precision is mainly limited by the implemented model complexity of the predictor. In first realizations of this idea, the behavior of the inverter was imitated by measuring the PWM signals and feeding them to an adaptive FIR filter, using its step responses as can-

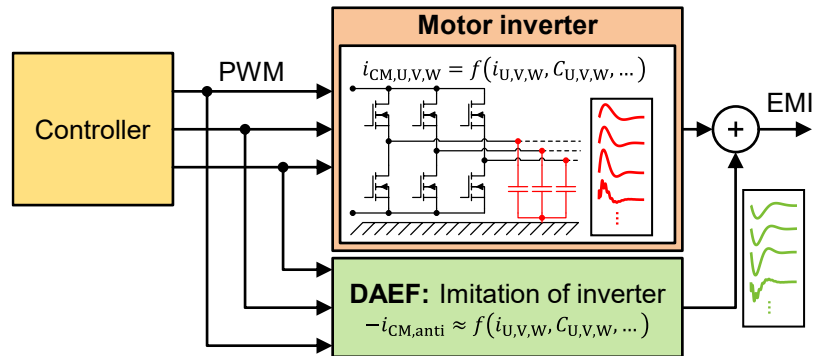


Figure 1: Concept of predictor DAEF for motor inverters.

cancellation pulses. This yielded a very broadband suppression of 26 dB up to 5 MHz, 19 dB at 10 MHz, and still 6 dB at 30 MHz [7] [8]. The approach was extended to suppress the turn-on and turn-off CM pulses of a DC-DC-converter with two separate cancellation waveforms, while exploring a more efficient signal generation through a PWM-triggered playback of prerecorded and optimized cancellation signals instead of an FIR filter [9]. However, these solutions still fall short in considering all possible influences on CM pulse generation in a drive train. For example, asymmetric parasitic capacitances of the motor may lead to different shapes per phase. Another crucial factor is the momentary phase current of the inverter, which can heavily influence the switching slope and CM EMI pulse of each phase [10]. To account for these influences, more sophisticated predictors are needed. One possible solution is the introduction of a physics-based model. As seen in [10], an analytic description of the switch node voltage depending on the momentary load current can significantly enhance CM EMI suppression of a single-phase inverter.

In this work, a data-driven solution is presented, where the CM pulses are separated from a single measurement of the inverter during operation. By distinguishing between the different pulses, each CM pulse can be assigned to, e.g., the responsible phase, switching direction, and momentary current. The separation takes advantage of the correlation between the PWM signals and the resulting disturbance pulses. With the separated disturbance pulses, the PWM-based DAEF can

be significantly improved. The optimal cancellation pulse can be selected from a database filled with the previously identified cancellation signals.

This paper begins by briefly describing possible deviations in motor inverter setups and their effect on CM EMI. After that, a generalized description of the signal separation method is given and the necessary steps to derive fitting cancellation signals are shown. The results are demonstrated with a 48 V GaN based motor inverter to isolate and compensate the influence of asymmetric parasitic capacitances.

2 CM EMI Generation in Motor Inverters

To analyze possible influences and asymmetries in a motor inverter, the CM disturbance mechanism is considered. A corresponding equivalent circuit for a motor with inverter and artificial network for measuring the conducted emissions is shown in Figure 2. The inverter connects the phases alternately to the DC voltage via three half-bridges to generate the sinusoidal current required for the drive. However, each switching event causes a steep voltage slope, driving pulse-shaped charge / discharge currents to ground via the parasitic capacitances of the phases. These occur primarily between the transistors and their grounded heatsink, cables and shielding, as well as the stator windings and motor housing. The individual pulses superimpose to form the measurable common-mode current, which runs back through the artificial networks and can be described in a simplified way as:

$$i_{CM}(t) = C_U \cdot \frac{dv_U(t)}{dt} + C_V \cdot \frac{dv_V(t)}{dt} + C_W \cdot \frac{dv_W(t)}{dt} \quad (1)$$

In real setups, the disturbance current is also shaped by line inductances and the artificial network components, resulting in sharp pulses, often with decaying oscillation. The driving voltage slopes $dv(t)/dt$ are directly caused by the PWM switching events. Previous DAEF approaches take advantage of this relation by mimicking the inverter: The PWM signals are measured and applied as ideal step signals to the input of an FIR filter, whose step responses are adapted to match the CM disturbance pulses similar to equation (1). With the ideal PWM signals exciting the step responses of both the inverter and the AEF, efficient reduction has been achieved for symmetric and idealized setups, using just one FIR filter for all three phases and their respective turn-on and turn-off pulses [7], [8].

However, the pulse shapes may vary. Geometric asymmetries in windings and heatsink connections lead to phase-specific coupling capacitances. Different turn-on and turn-off gate resistors of the transistors affect the rising and falling slopes. Another factor is the time-varying load current, which determines the commutation behavior of the phase voltages and thus directly their time derivative [10]. Identifying suitable cancellation signals therefore requires a separate analysis of the pulses. Since all disturbance pulses are continuously overlapping during operation, the pulses superimpose into a complex composite signal, making direct separation of the signals impossible.

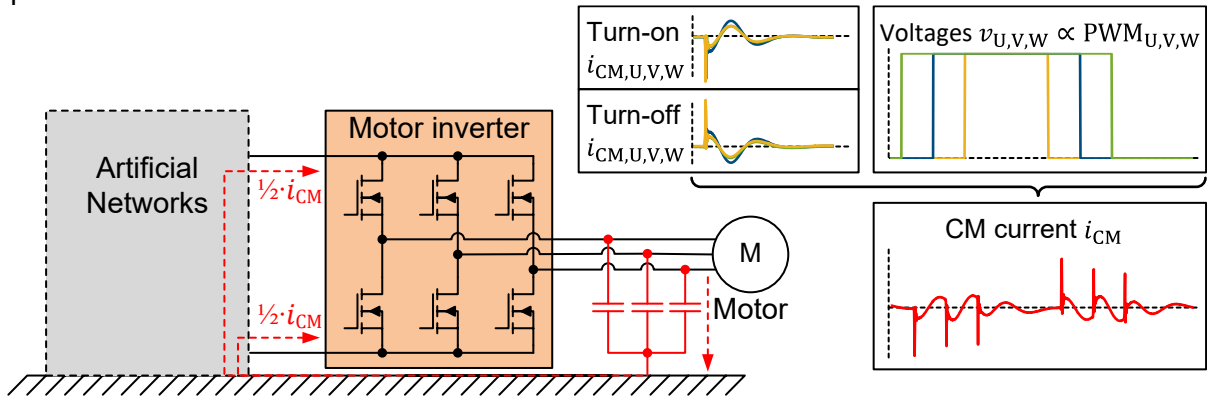


Figure 2: Fundamental CM EMI mechanism in motor inverter systems.

3 Separation and Reconstruction of CM EMI Pulses in a Motor Inverter

3.1 Disturbance Signal Modeling through Convolution

To reconstruct individual disturbance pulses from a measured CM signal, which contains the superposition of pulses caused by different PWM sources (e.g., the turn-on pulse for phase U), a method is developed that utilizes their correlation to the PWM switching signals. The disturbance pulses are interpreted as impulse responses $h_k(t)$ to the turn-on and turn-off switching edges of the PWM signals. For this, input signals $u_k(t)$ are derived as a series of Dirac pulses $\delta(t)$ at the respective turn-on and turn-off times of the PWM. Consequently, the measurable disturbance current results from the superposition of convolutions of the switching edges with the unknown impulse responses:

$$i_{CM}(t) = \sum_{k=1}^K u_k(t) * h_k(t) \quad (2)$$

3.2 Separation of the PWM Signals

The number K of different input signals and respective impulse responses depends on the model's desired level of detail: In the simplest configuration, a single pulse shape $h_1(t)$ is used. The corresponding input $u_1(t)$ would consist of positive Dirac pulses $\delta(t)$ at the turn-on times and negative Dirac pulses $-\delta(t)$ at the turn-off times of all PWM signals combined. This actually represents the approach used in previous publications [7], [8], where a single FIR filter was used to produce fitting cancellation pulses triggered by the edges of a combined PWM signal as stated in section 2. Increasing the complexity to distinguish between turn-on and turn-off edges with respective pulses $h_1(t)$ and $h_2(t)$ allows for the mapping of different gate resistors. A separation by the three phases can help identify asymmetric parasitic capacitances in the inverter topology. To analyze phase current dependency, specific impulse responses may be assumed for different current levels. As some features might prove significant in one system compared to another, the level of separation must be chosen accordingly. With this concept applied to the example in Figure 2, the displayed CM current may be modeled as a convolution of PWM signal edges with the respective pulse waveforms on the left, for a total of $K = 6$ convolutions.

3.3 Discretization and Vectorial Formulation

With a synchronous measurement of the CM EMI and the PWM signals, the convolution (2) may be discretized, which can also be expressed as a vector equation. For this, the impulse responses are concatenated in vector $\mathbf{h}$:

$$\mathbf{h} = [\mathbf{h}_1 \quad \mathbf{h}_2 \quad \dots \quad \mathbf{h}_K]^T = [h_1[0] \quad \dots \quad h_1[M-1] \quad h_2[0] \quad \dots \quad h_K[M-1]]^T \quad (3)$$

Where the impulse response length M must be chosen sufficiently to cover the duration of one decaying disturbance pulse. Next, the discrete convolution matrix $\mathbf{U}$ must be defined, so that a multiplication with the vector $\mathbf{h}$ of the unknown impulse responses yields the total disturbance vector $\mathbf{i}_{CM}$ defined by the measured series $i_{CM}[n]$:

$$\mathbf{U} \cdot \mathbf{h} = \mathbf{i}_{CM} \quad (4)$$

$\mathbf{U} = [\mathbf{U}_1 \quad \mathbf{U}_2 \quad \dots \quad \mathbf{U}_K]$ consists of K horizontally concatenated convolution matrices $\mathbf{U}_k$ from the respective input signals $u_k[n]$. Each convolution matrix has N rows for the recorded samples of the total disturbance vector $\mathbf{i}_{CM}$ and M columns for the length of its respective impulse response $\mathbf{h}_k$. This results in matrix $\mathbf{U}$ containing diagonals of ones with length M , positioned according to the PWM edge time (row) and the corresponding impulse response (column). Consequently, a switching event activates the associated impulse response, which contributes to the total disturbance for the next M samples.

3.4 Signal Reconstruction through Inverse Convolution

Each row of the matrix-vector multiplication (4) maps the superposition of the unknown impulse responses $\mathbf{h}$ to the total disturbance current i_{CM} . Inverting the matrix reverses the convolution, allowing the reconstruction of the impulse responses from the measured disturbance:

$$\mathbf{h} = \mathbf{U}^{-1} \cdot \mathbf{i}_{CM} \quad (5)$$

For a real-world application, extending the recording duration beyond the minimum number of samples reduces the impact of noise. Furthermore, varying time delays between the PWM edges are essential to avoid linearly dependent columns in the matrix. Motor inverters inherently fulfill this condition through the modulation-based variation of pulse widths.

Long measurement signals with uncorrelated switching times result in an overdetermined system for which no direct inverse $\mathbf{U}^{-1}$ exists. Since measurement noise also makes an exact solution impossible, a least squares (LS) optimization is applied. This determines the solution vector $\mathbf{h}$ by minimizing the quadratic error between the calculated convolution and the measured disturbance:

$$\min_{\mathbf{h}} \|\mathbf{U} \cdot \mathbf{h} - \mathbf{i}_{CM}\|^2 \quad (6)$$

As a simple example, a configuration with just $K = 2$ impulse responses $h_1[n], h_2[n]$ with a short length of $M = 3$ is considered. During the measurement length of $N = 9$, the first PWM input signal $u_1[n]$ switches at times $n = 1$ and $n = 5$, the second input $u_2[n]$ at $n = 2$ and $n = 7$. With the sum signal $i_{CM}[n]$ observed over the measurement window, the vectors are established as:

$$\mathbf{u}_1 = [0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0]^T, \mathbf{u}_2 = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0]^T \\ \mathbf{i}_{CM} = [-1 \ 4 \ 4 \ 1 \ -1 \ 2 \ 3 \ 3 \ 1]^T, \mathbf{h} = [h_1[0] \ h_1[1] \ h_1[2] \ h_2[0] \ h_2[1] \ h_2[2]]^T \quad (7)$$

The convolution matrix $\mathbf{U}$ is constructed from the two input signals and equation (4) is rewritten, which can be solved like equation (6):

$$\mathbf{U} \cdot \mathbf{h} = \mathbf{i}_{CM} \Leftrightarrow \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} h_1[0] \\ h_1[1] \\ h_1[2] \\ h_2[0] \\ h_2[1] \\ h_2[2] \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 4 \\ 1 \\ -1 \\ 2 \\ 3 \\ 3 \\ 1 \end{bmatrix} \xrightarrow{\text{LS}} \mathbf{h}_1 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{h}_2 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \quad (8)$$

In the example, the columns are clearly linearly independent, as the Diracs in $u_1[n]$ and $u_2[n]$ have varying time delays to each other. Thus, the system of linear equations can be solved.

With the disturbance pulses separated, the CM EMI dependencies on factors such as the motor phase, switching direction and with an extension the phase current can be analyzed. Furthermore, the signals can be used to determine fitting cancellation pulses for precise EMI suppression.

4 Extended Use for Active EMI Cancellation

Based on the separated disturbance pulses, the PWM-based DAEF approach presented in the introduction can be enhanced. For this, individual cancellation waveforms are derived to efficiently suppress the identified pulses. These can be stored in a digital memory and injected via a digital-to-analog converter, an amplifier, and a passive coupling circuit during operation of the inverter for precise cancellation.

4.1 Derivation of the Cancellation Signals

The block diagram shown in Figure 3 illustrates the active filter integration. Each inverter leg produces disturbance pulses caused by the PWM transients, modeled through a convolution and summation as described in section 3.1. In parallel, the DAEF plays back appropriate cancellation signals based on the switching edges. These traverse the cancellation path $S(z)$ as a composite

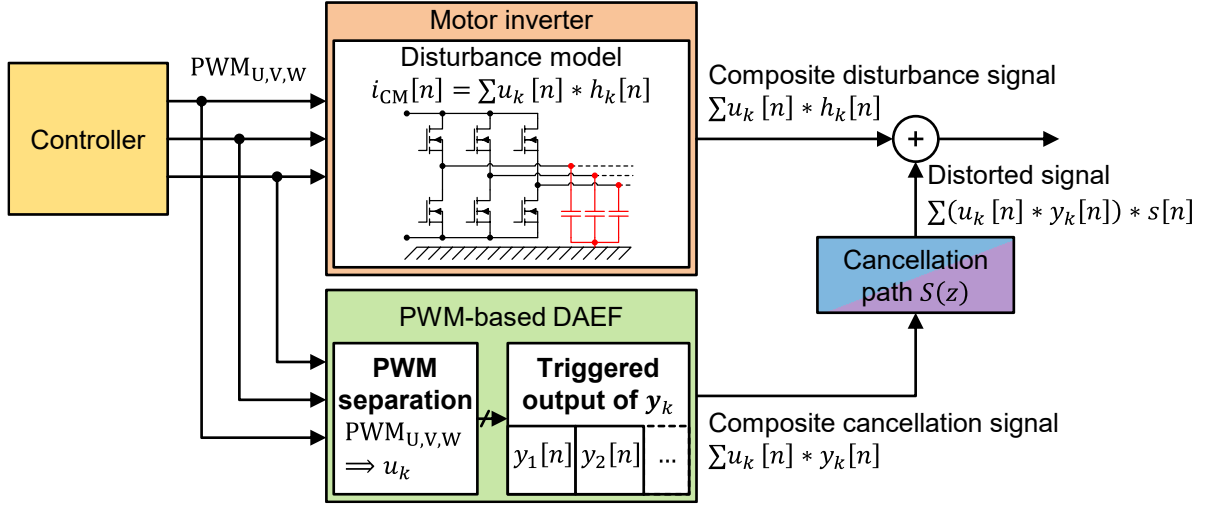


Figure 3: Block diagram of PWM-based DAEF implementation.

signal and cancel the system disturbance at the summation point through superposition. With an ideal cancellation path, the impulse responses $-h_k[n]$ determined in section 3.4 could be used directly as cancellation signals $y_k[n]$. However, since the real setup includes the digital signal processor, a digital-to-analog converter, an amplifier, and a passive coupling circuit [8], the distortion introduced by the cancellation path must be considered. This can be described as a convolution with the secondary path impulse response $s[n]$, where the result should cancel $h_k[n]$:

$$y_k[n] * s[n] \stackrel{!}{=} -h_k[n] \quad (9)$$

To derive the necessary cancellation pulses, the convolution (9) must be inverted again. In analogy to sections 3.3 and 3.4, a vector equation with the convolution matrix $\mathbf{S}$ of $s[n]$ is established for each signal $h_k[n]$ and the LS optimization is written:

$$\begin{aligned} \mathbf{S} \cdot \mathbf{y}_k &\stackrel{!}{=} -\mathbf{h}_k \Leftrightarrow \mathbf{y}_k = -\mathbf{S}^{-1} \cdot \mathbf{h}_k \\ &\stackrel{\text{LS}}{\Rightarrow} \min_{\mathbf{y}_k} \|\mathbf{S} \cdot \mathbf{y}_k + \mathbf{h}_k\|^2 \end{aligned} \quad (10)$$

However, this inverse convolution with the non-ideal impulse response $s[n]$ is an ill-posed LS problem, as frequency attenuations in the cancellation path results in an ill-conditioned matrix $\mathbf{S}$. For example, a capacitive injection as used in [8] blocks DC components and leads to infinite solutions with arbitrary DC values. Therefore, a single solution must be enforced. For this, ridge regression [11] is used, adding a penalty for large signal amplitudes to the optimization problem:

$$\min_{\mathbf{y}} \|\mathbf{S} \cdot \mathbf{y} + \mathbf{h}\|^2 \xrightarrow{\text{Ridge-R.}} \min_{\mathbf{y}} (\|\mathbf{S} \cdot \mathbf{y} + \mathbf{h}\|^2 + \lambda \|\mathbf{y}\|^2) \quad (11)$$

Where the regularization factor λ controls the penalty applied to large values in the solution vector $\mathbf{y}$. Ridge regression thus enables a numerically stable inverse convolution to identify the cancellation pulses for the DAEF.

4.2 Cancellation Signal Generation

After successful calculation of the cancellation signals through inverse convolution, the waveforms are stored in the cancellation system. To generate the signals at the output during operation of the inverter, the PWM transients are used as triggers to play back the cancellation pulses. For this, the DAEF must separate the PWM switching edges in the same scheme as specified in section 3.2. Like the example of Figure 2, they may be dissected by the switching direction (on / off) and the phase (U, V, W), leading to $K = 6$ different triggers. If the current dependency of the pulses is to be considered, the phase current must be introduced as another input. For example, a lookup table which is indexed by the measured phase currents can select the fitting waveform for the momentary current of each phase. By this method, virtually any measurable factor affecting the CM pulses can be considered in the cancellation signal generation.

5 Exemplary Application

5.1 GaN-Based Motor Inverter as Demonstrator Setup

The developed method is demonstrated for a 48 V GaN-based motor inverter depicted in Figure 4. The setup consists of a DC voltage source, two CISPR-25 artificial networks with a CM combiner, the motor inverter, and a RL motor emulation load. The CM EMI is measured at the artificial networks with an EMI test receiver. For the interfaces of the active EMI filter, capacitive injection and sensing circuits with 22 nF and 15 nF respectively are selected in analogy to [8]. The AEF software is running on a Red Pitaya STEMLab 125-14 FPGA, which also generates the PWM signals for the inverter. In other setups, these functions can be split across two components where the AEF FPGA measures the PWM signals generated by the controller. For cancellation, the FPGA holds the required cancellation waveforms in internal block-RAM storage cells, which can easily be triggered by the PWM edges and generate the waveforms at the output digital-to-analog converter. To increase the maximum cancellation current, amplifiers are used in the capacitive injection circuit. The sample rate of the FPGA is 125 MSPS.

For this particular setup, the current dependency is neglected, since the GaN switches of the inverter have a very low dead time, which makes the current irrelevant [12]. To demonstrate the effectiveness of the signal separation from phase to phase, asymmetric ground capacitances C_{GND} are added to the emulation load with 1,100 pF for phase U and 680 pF for phases V and W. Additionally, the turn-on and turn-off shall be handled separately. Therefore, the edges of the three PWM signals are dissected by their phase and switch direction for $K = 6$ pulse shapes.

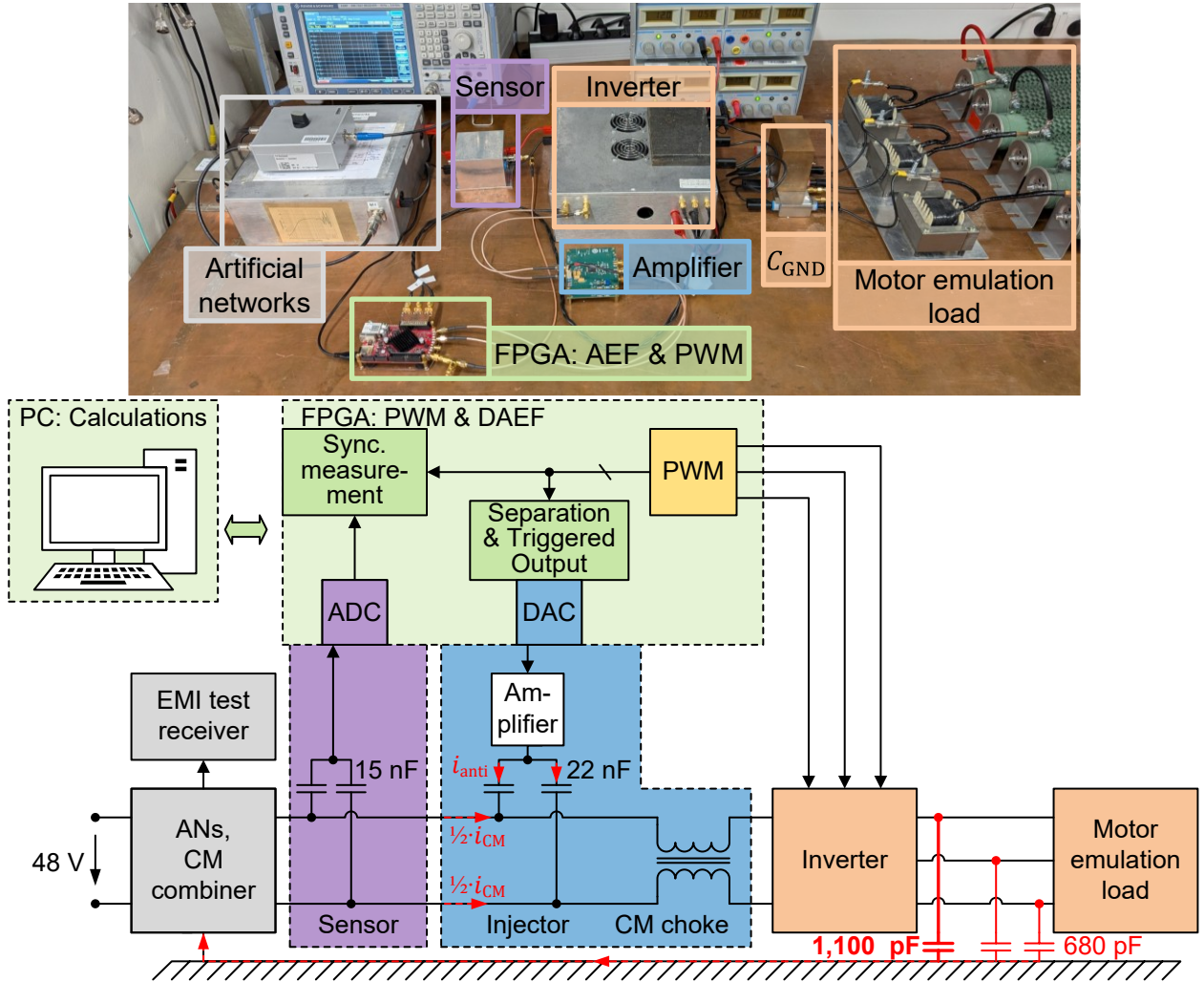


Figure 4: Measurement setup and block diagram.

5.2 Signal Separation

In the setup, the FPGA records a synchronous measurement of the PWM and disturbance signals as required for the signal separation of section 3. This dataset is sent to a PC for calculation of equation (6) in MATLAB. As specified, the algorithm shall distinguish between each phase and the turn-on and turn-off pulses for a total of $K = 6$ separable signals. For the calculation, one 50 Hz period with $N = 2,500,000$ samples is recorded. A length of $M = 1,000$ is selected as a sufficient length for each impulse response, which translates to $8 \mu\text{s}$ with the sample time of 8 ns. This leads to a dimension of $2,500,000 \times 6,000$ for matrix $\mathbf{U}$. The results of the decomposition are shown in Figure 5. As a reference for each pulse, a measurement during DC/DC operation is shown in grey where only one phase is switching with the others turned off, not requiring any signal separation. This mode however is only viable with an equivalent RL load, as a real motor would not have enough DC resistance and short the phase. While it is impossible to separate the different pulses in the measured CM EMI by hand, the algorithm successfully reconstructs the 6 individual disturbance pulses, closely matching the reference curves from the DC/DC operation. The effect of the higher capacitance connected to phase U can clearly be seen in the larger resonance.

5.3 Cancellation Signal Calculation

In the next step as defined in section 4, the identified disturbance signals must be deconvolved with the cancellation path impulse response $s[n]$ to reconstruct the necessary cancellation signals. For this, the FPGA injects a series of impulses into the cancellation path while the inverter is turned off to measure and identify the impulse response. With this, equation (11) is computed on the PC. Figure 6 depicts the identified cancellation path impulse response and the phase U turn-on

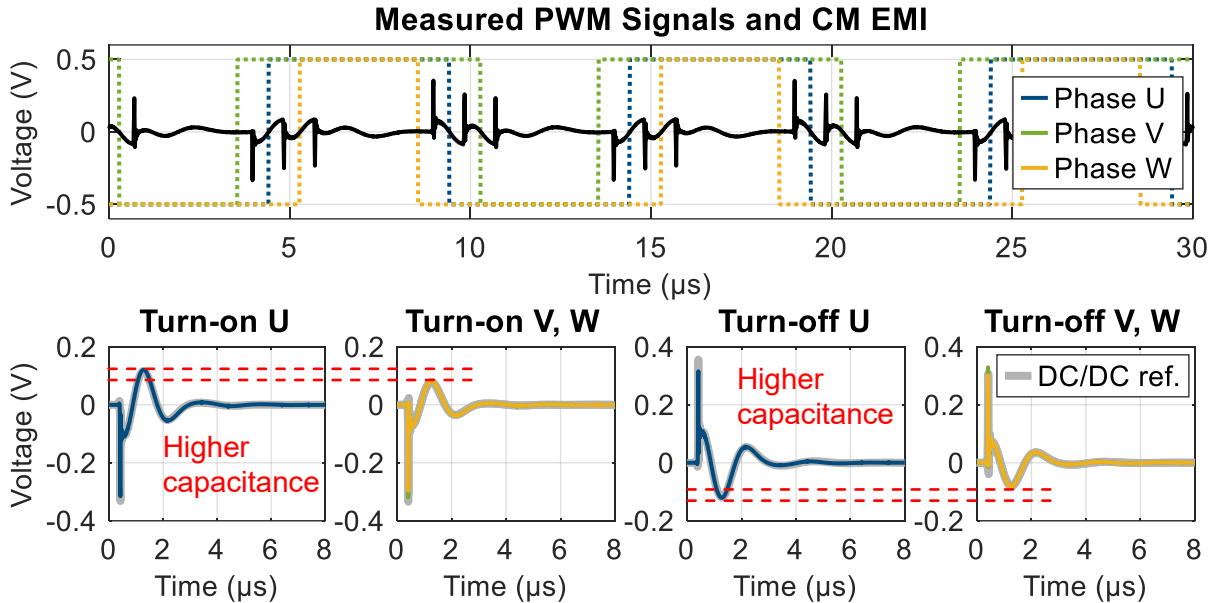


Figure 5: Measured PWM signals and CM EMI and resulting separated turn-on / turn-off pulses.

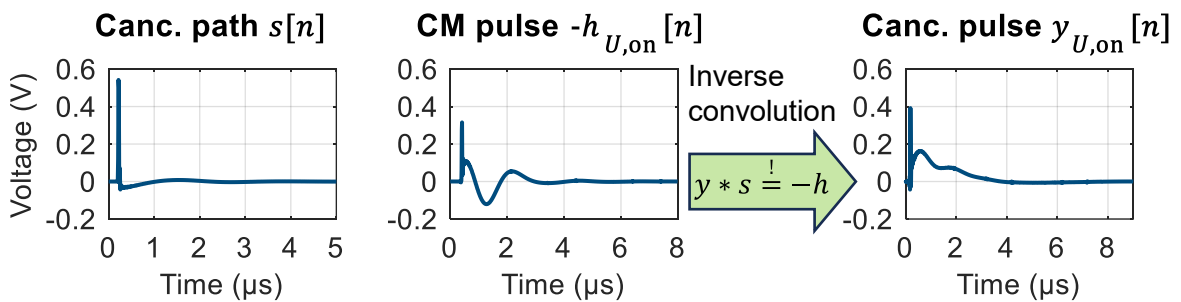


Figure 6: Identified cancellation path impulse response, target CM pulse and cancellation pulse resulting from inverse convolution.

waveform both before and after deconvolution. For $s[n]$, a delayed sharp impulse with a resonance can be observed. Because the cancellation signal has to compensate for that resonance, its length of $M_y = 1,200$ is selected longer than the length of $M = 1,000$ of the original noise pulse. The process shown exemplary for one waveform in Figure 6 is done for all 6 disturbance pulses, which are then uploaded to the FPGA for cancellation.

5.4 Cancellation Results

With all six cancellation waveforms stored in the FPGA, they can be played back triggered by the PWM edges to cancel out the CM EMI measured at the artificial networks. The results are shown in Figure 7. Here, the broadband cancellation effect can clearly be seen with around 25 dB at lower frequencies of 2 MHz, 20 dB at 5 MHz and still 8 dB at 20 MHz. To highlight the importance of considering the asymmetric capacitance in the cancellation signals, an additional measurement is done where all the identified impulse responses are averaged into a singular cancellation pulse. This represents the previous implementation as stated in section 3.2 with just one shape for all occurring noise pulses, yielding a significantly lower suppression, especially at lower frequencies. This is plausible as Figure 5 shows that the pulses mostly deviate in their low-frequency behavior, while the fast initial transients are shaped very similarly.

6 Conclusion and Outlook

In this work, a novel method for effective separation of individual disturbance pulses from the measurable CM EMI signal of a motor inverter has been presented. For this, a discrete convolution model based on the correlation between the PWM signals and CM pulses was developed, which can be inverted to identify the disturbance waveforms. In an extension, fitting cancellation pulses were derived to consider different CM pulse shapes in active EMI filters. The approach was demonstrated to compensate the CM noise on the supply lines in a 48 V GaN motor inverter with asymmetric parasitic capacitances, yielding significantly better results when considering these differences with unique cancellation pulses for each phase and turn-on / turn-off transient. As described, the generalized method can also be used to separate CM pulse shapes by other factors, such as the momentary phase current. Therefore, the concept can be applied to disturbances of motor inverters with a significant current dependency. Additionally, the second step of calculating fitting cancellation signals could be enhanced: At the current state, each cancellation waveform is a pulse that has to return to zero at the final sample. However, the AEF output voltage should actually imitate the trapezoidal shape of the phase voltages, as both are capacitively coupled. Therefore, trapezoidal signals with arbitrary slopes would pose the ideal cancellation signal, which could be implemented in future research.

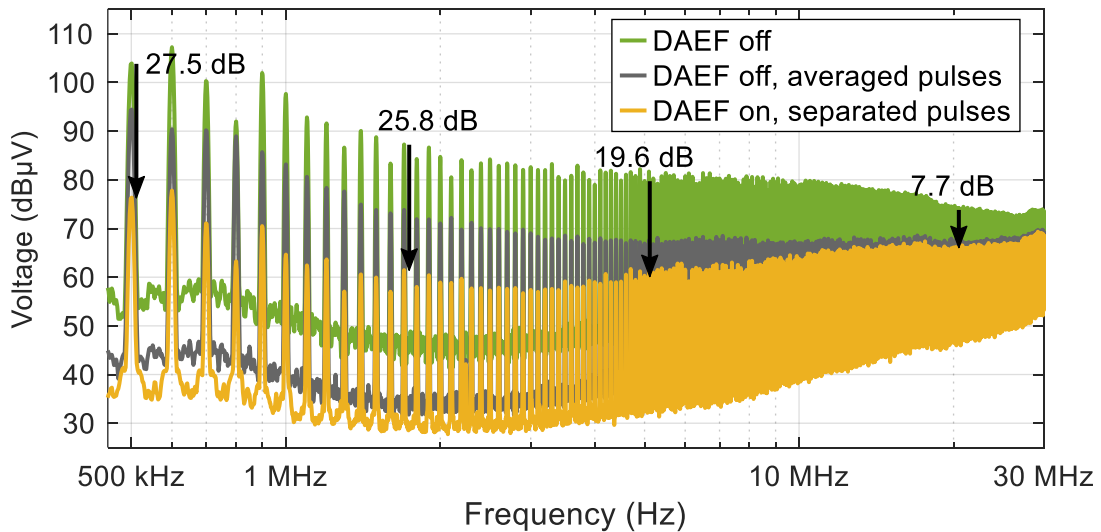


Figure 7: CM EMI spectra with AEF disabled and enabled using averaged and separate pulses.

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